

ASHAR AYUB

Lead Software Engineer · Distributed Systems, Real-Time Platforms, Production LLM Integration

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SUMMARY

Lead Full Stack Engineer with 5+ years designing scalable backend infrastructure, real-time communication systems, and dynamic client-side applications. Highly proficient in AI-augmented development (Cursor IDE, Claude Code, AntiGravity, Model Context Protocol), utilizing these tools to accelerate cross-stack delivery cycles, rapidly adopt new frameworks, and significantly reduce time-to-market. Integrate LLMs into production systems and lead architectural review of both human-authored and AI-generated code. Mentor junior engineers and hold accountability for correctness in regulated domains including clinical and management platforms.

CORE SKILLS

Architecture & Cloud AWS (EC2, RDS, S3, Load Balancers, multi-region), Azure, GCP, Cloudflare, MinIO, Nginx, Ubuntu/VPS, Laravel Forge, CI/CD (GitHub Actions), zero-downtime deployment.

Backend PHP/Laravel, Node.js, NestJS, Python/FastAPI, REST API, GraphQL, PHP.

Real-Time & Data WebSockets, Coturn (TURN/STUN), Redis, MySQL, PostgreSQL, MongoDB, query tuning & indexing, caching, queues, Crons.

AI in Production LLM integration (FastAPI, vector DBs), agentic workflows, Model Context Protocol (MCP), AI-generated code review & architectural governance.

Frontend & Mobile Vue/Inertia.js, React, Angular, Ionic/Capacitor, React Native, Tailwind CSS, Kotlin, Java.

PROFESSIONAL EXPERIENCE

Lead Developer — KLABS Tech Solutions Jun 2021 – Present · Karachi, PK

- **Architectural governance:** Led server-side architecture across concurrent client engagements; conduct design and code review of both human-authored and AI-generated components, owning correctness, security, and production accountability.
- **Mentorship:** Mentored 5+ junior engineers across backend, mobile, and development work; established review standards and delivery practices for the team.
- **Platform engineering:** Architect RESTful services in PHP/Laravel and Node.js; drive database performance through query tuning, indexing, and caching strategies — [add a measured before/after here].
- **Real-time systems:** Build instant messaging, live broadcast feeds, and push notification infrastructure on PHP/Laravel, NodeJS, WebSockets, and Redis backplanes serving 5+ concurrent connections.

Selected client engagements

CODOT — Distributed Digital Signage Platform · Full Stack Engineer & Cloud Architect[2024 – 2026]

- **Multi-region disaster recovery:** Led a failover from AWS Bahrain to AWS Mumbai — full infrastructure reprovisioning, database and S3 migration — after regional data centres were destroyed during the Iran conflict. Service restored in a few days with 200+ devices reconnected.
- **Sovereign storage workaround (Oman):** Designed and deployed custom MinIO object storage to serve the Oman region, where native AWS bucket support was unavailable, keeping media delivery uninterrupted across endpoints.
- **P2P streaming architecture:** Built a Node.js WebSocket server paired with a custom Coturn (TURN/STUN) deployment for low-latency P2P live streaming and bidirectional telemetry between the management dashboard and distributed player devices — [X ms] median latency.
- **Native playback client:** Delivered the hybrid signage player, then rewrote it natively in Kotlin to gain hardware acceleration, improving playback.

- **Control plane:** Designed the backend APIs and administrative portal managing device state, ad workflows, and real-time playback monitoring at fleet scale.

RCSI AI Simulator — Clinical Training Platform · Lead Full Stack & AI Integration Engineer[2024]

- **Production LLM integration:** Worked alongside AI specialists to embed LLMs into a live clinical simulation environment driving real-time simulated patient dialogue and automated triage scoring for medical practitioners.
- **State-intensive backend:** Engineered the core Laravel infrastructure and tuned SQL schemas to sustain long-running, state-heavy simulation sessions for multiple concurrent candidates.
- **Frontend leadership:** Guided the React/Inertia.js SPA development with code review and mentorship; configured a Laravel Nova portal giving non-technical administrators self-serve control of candidates, scenarios, and assessment analytics under zero-downtime deploys.

AL MSAYYAN (Eskan Bank) — Land & Plot Management Platform · Senior Software Engineer[2024]

- **Interactive geospatial system:** Built a live land visualisation and plot management platform using Leaflet.js with custom map overlays and dynamic interactive polygons across 250+ residential plots.
- **Real-time state sync:** Implemented live polling to synchronise plot status and colour-coded availability, with Laravel Turbolinks for partial page loads; extended the admin dashboard so non-technical staff manage inventory and map coordinates without developer involvement.

Jamaheer (Bahrain Premier League) — Football Analytics Platform · Backend Development Lead[2026]

- **Infrastructure & launch:** Provisioned resilient VPS infrastructure and domain routing for a high-traffic live-score and analytics platform serving concurrent users at peak matchday load.
- **Real-time match architecture:** Designed backend systems for live match tracking and tournament data, integrating with cross-functional mobile teams; delivered squad lineup builder, live match control, and statistics generation using AI-assisted development to compress the build cycle to 8 weeks.

Additional engagements · Vigilance Consulting · King Hamad Global Center (Bahrain) · Raed, Injaz Academy, LAX[2021 – 2026]

- Led Laravel/Statamic CMS architecture and migrations for two client platforms — converting static and legacy Vue/Laravel sites into self-serve, no-code content environments with zero data loss; owned production deployment via Laravel Forge.
- Delivered cross-platform EdTech and engagement mobile apps (Ionic, Angular, Laravel) for Middle Eastern markets, including enrollment, assessment, and scheduled push-notification pipelines.

Freelance Software Architect (concurrent) 2020 – Present

LetsUpDoc — Telemedicine Platform · End-to-End Architect[2023 – Present]

- **Full platform ownership:** Directed the complete lifecycle — patient portals, real-time video consultation, diagnostic lab results, pharmacy inquiry — across web and mobile, published to the App Store and Google Play.
- **LLM clinical triage:** Architected FastAPI-based LLM integration embedded in the mobile interface for sub-second triage and workflow evaluation, with clinical review gating on every automated output.
- **Infrastructure:** Provisioned and secured Ubuntu VPS environments (Nginx, SSL, DNS) with zero-downtime GitHub Actions pipelines on PHP Swoole / Laravel Octane.

LetsUpMart — E-Commerce Platform · Platform Engineer[2024 – Present]

- **Catalog automation at scale:** Built Python scrapers and ETL pipelines ingesting 1000+ products from external sources, with an LLM post-processing stage that rewrites descriptions for uniqueness and SEO — replacing 10+hours/week of manual cataloging.
- **Custom commerce architecture:** Heavily extended a Bagisto/Laravel base, re-engineering native admin workflows and building new operational pipelines for bespoke order management; led technical SEO and performance work improving [PageSpeed / indexing metric].

EDUCATION

BSc Computer Science — NED University, Karachi · CGPA 3.512017 – 2020